



Air University
Final Examination: Fall 2025

Student ID: _____

Paper Type: Subjective
Note: To be solved on Answer book only

Time Allowed: 3 Hours
Total Marks : 100

Subject: Artificial Intelligence

Course Code: CS3304

Class: Fall-23

Section(s): A

Special Instructions:

Total Time Allowed: 03 Hrs.

FM Name: Muhammad Hussain

FM Signature:

INSTRUCTIONS

- **Rough work or writing on question paper will be considered as use of unfair means.**
- **Tables/calculators allowed/not allowed.**

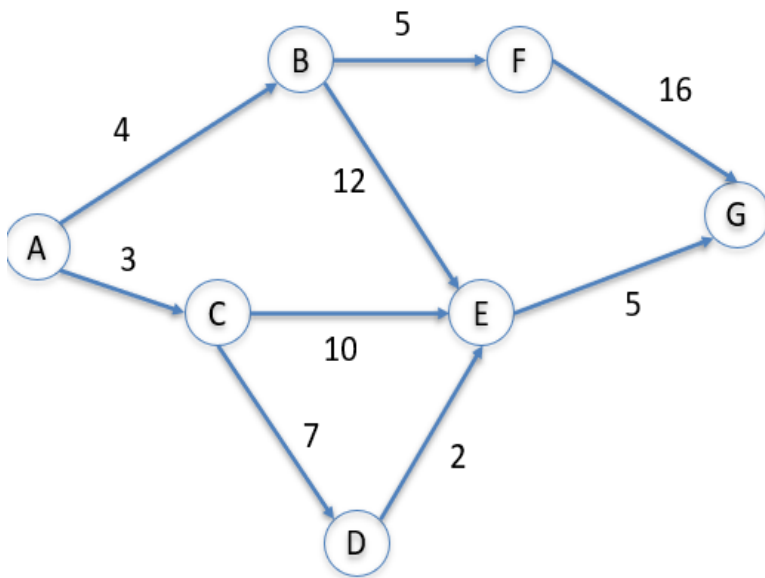
Q1. (CLO-01 & PLO-01: Explain artificial intelligence, terminologies, its characteristics and its application areas.) Marks (20)

- Define** Artificial Intelligence (AI).
- Define** an Agent in the context of AI.
- Define** Utility Based Agent also explain it with the help of a diagram.
- Define the environment and explain any three types of environments.

Q2. (CLO-02 & PLO-02: Solving search problems using various uninformed and informed search algorithms. Apply local search algorithms like Genetic Algorithm (GA) on optimization problems and perform Minimax search on games.) Marks (20)

- What is Informed Search in Artificial Intelligence? Explain how heuristic functions guide the search process.
- Describe the working of Greedy Best-First Search.
Also mention one advantage and one limitation of this algorithm
- Explain the A* search algorithm?
How does A* combine path cost and heuristic information?
- Consider the following search problem:
A graph is given where the start node is S and the goal node is G.
Graph, heuristic values $h(n)$ and path costs are shown below:

Note: Show step by step working



Node	h(n)
A	14
B	12
C	11
D	6
E	4
F	11
G	0

Q3. (CLO-03 & PLO-03: Implementing various machine learning concepts including data handling, attribute subset selection, feature extraction, regression and classification) Marks (30)

- Define Machine Learning and explain how it is different from traditional programming.
- Define Supervised and Unsupervised Learning.
- Define how K-Nearest Neighbor (KNN) algorithm works.
- Consider the following training data:

Points	Feature 1 (X)	Feature 2 (Y)	Class
A	2	4	Red
B	4	6	Red
C	6	8	Blue
D	8	10	Blue

A new data point P (5,6) is given. Classify point P using K = 3.

Q4. (CLO-03 & PLO-03: Implementing various machine learning concepts including data handling, attribute subset selection, feature extraction, regression and classification) Marks (30)

- What is a Confusion Matrix in Machine Learning and Why it is important for evaluating a classification model?
- A binary classification model produced the following results on a test dataset:

Predicted	Actual		
		Positive	Negative
	Positive	50	10
	Negative	5	35

Calculate the following performance measures:

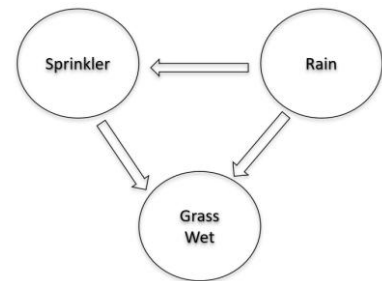
- Accuracy
- Sensitivity
- Specificity

c) Define a Bayesian Network and explain why it is useful for reasoning under uncertainty.

d) Consider a backyard where the wetness of grass (**W**) depends on **Rain (R)** and a **Sprinkler (S)**.

- The probability of rain is $P(R) = 0.3$
- If it rains, the sprinkler turns on with probability $P(S | R) = 0.01$.
- If it does not rain, the sprinkler turns on with probability $P(S | \neg R) = 0.40$.

The probability of the grass being wet is:



Conditional probabilities:

Rain	Sprinkler	P (W = True R, S)
T	T	0.95
T	F	0.80
F	T	0.70
F	F	0.05

- Calculate the probability that the grass is wet $P(W)$.